**Phase 1 of 6**: Core language foundations · lesson 1 of 37

# C++ Basics

Start here if C++ is new or rusty: how a program is built, variables and types, operators and control flow, functions and passing arguments, and the tools that catch mistakes. Pointers and References is the next lesson.

## How a C++ program is built




Compile error: a problem inside one file. Linker error: declared and called but never defined.

Three stages. Know which one an error comes from.

C++ is **compiled** to machine code before it runs: fast, but it trusts you not to make memory mistakes.

main.cpp

```cpp
#include <iostream>          // paste in the declarations for std::cout

int main() {                 // execution starts here
    std::cout << "Hello, C++\n";
    return 0;                // 0 means success
}
```

1. **Preprocess**: `#include` literally pastes header files in; `#define` does text substitution.
2. **Compile**: each `.cpp` file becomes an *object file* on its own.
3. **Link**: the linker joins object files and libraries into one executable and resolves calls between files.

KEY POINT "Compile error vs linker error?" A compile error is a syntax or type problem inside one file. A linker error ("undefined reference") means a function was declared and called but never defined anywhere you linked.

## Variables and types

Every variable has a **type** that fixes how much memory it uses and what you may do with it.

| TYPE | HOLDS | TYPICAL SIZE |
|---|---|---|
| `bool` | `true` / `false` | 1 byte |
| `char` | a character / small integer | 1 byte (always, by definition) |
| `int` | whole number | 4 bytes |
| `long long` | bigger whole number | 8 bytes |
| `float` / `double` | decimal number | 4 / 8 bytes |

Only `char` has a fixed size. When size matters use `<cstdint>`: `std::int32_t`, `std::uint64_t`.

### Initialization: always give a variable a value

```cpp
// BAD: indeterminate value, reading it is undefined behavior
int a;
int b = 5;       // copy initialization
int c(5);        // direct initialization
int d{5};        // brace (uniform) initialization: preferred
int e{};         // empty braces: zero-initialized, e == 0
int f{3.7};      // ERROR: braces forbid narrowing double → int
auto g = 5;      // g is int: the compiler deduces the type
const int h = 9;// cannot be changed after initialization
```

> UNDEFINED BEHAVIOR An uninitialized local holds garbage. Reading it is **UB**: junk, a crash, or "works" until it does not.

### The signed/unsigned trap

`unsigned` wraps instead of going negative, and a mixed comparison converts the signed side to unsigned.

## Operators and control flow

```
7 / 2      // 3     integer division truncates
7 % 2      // 1     remainder
// 3.5 one double operand makes the whole thing floating point
7 / 2.0

// && and || short-circuit: right side is skipped if not needed
if (x > 0 && y > 0) { }
for (int i = 0; i < 3; ++i) { }
for (int v : vec) { }     // range-based for: visits every element
while (cond) { }
// forget `break` and it falls through
switch (n) { case 1: /* ... */ break; default: break; }
```

> CLASSIC BUG `if (x = 5)` *assigns* 5 to x and is always true. You meant `==`. The compiler warns with `-Wall`.

## Functions and passing arguments

Declare before use; the header holds declarations, the .cpp the definitions.

```
int add(int a, int b);                 // declaration
int add(int a, int b) { return a + b; }// definition
// overload: same name, different parameter types
double add(double a, double b);
void greet(std::string who = "world"); // default argument
```

### Three ways to pass a parameter

```
void byValue(int n)        { n++; }          // works on a COPY
void byRef(int& n)         { n++; }          // works on the ORIGINAL
// original, read-only, no copy
void byConstRef(const std::string& s) { }
```

| YOU WANT TO... | USE |
| --- | --- |
| read a small value (int, double, char, pointer) | by value: `int n` |
| read a large object without copying it | by const reference: `const std::string& s` |
| modify the caller's variable | by reference: `int& n` |
| accept "no object" as a valid argument | by pointer: `Widget* w`, and test it for `nullptr` |
| pass a run of elements | `std::span<int>` (C++20) or `const std::vector<int>&`, not a pointer plus a count |
| get a result back | return it by value; big objects such as `std::vector` are moved, not copied, so no out-parameter is needed |

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## Tooling: warnings, sanitizers, static analysis

C++ trusts you, so let tools check what the language will not. Use them in layers, cheapest first.

| LAYER | WHAT IT CATCHES | HOW |
| --- | --- | --- |
| Compiler warnings | many beginner mistakes at once: uninitialized variables, signed/unsigned comparisons, missing returns | `-Wall -Wextra`, and `-Werror` in CI |
| Sanitizers | what only shows at run time: out-of-bounds, use after free, leaks, undefined behavior, data races | `-fsanitize=address,undefined` (and `-fsanitize=thread` for threads) |
| Static analyzer | rule violations, null dereferences, leaks and dead code found by reading the source, without running it | `clang-tidy`, `cppcheck` |
| Formatter | style, so reviews are about logic | `clang-format` |

**A CLEAN REPORT IS NOT A PROOF** Analyzers give false positives and miss things, and a sanitizer only checks the paths your test actually ran. They reduce risk; they do not replace design, review and tests.

**CODING STANDARDS** Safety-critical projects (cars, medical devices) add written rules such as MISRA C++ or the AUTOSAR C++14 guidelines, checked automatically. The common themes are the ones in this course: initialize everything, no C-style casts, no naked `new` / `delete`, no silent narrowing, and no `using namespace` in headers.

## Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

| SECTION | BOOK | PART OF THE BOOK | PDF PAGES |
|---|---|---|---|
| **How a C++ program is built** | C++ Primer | 2.6 Defining our own data structures | 101 to 109 |
| | C++ Primer | 3.1 Namespace using declarations | 111 to 112 |
| | C++ Primer | 18.2 Namespaces | 814 to 830 |
| | Programming: Principles and Practice | 7.2 to 7.3 Declarations and definitions; scope | 194 to 202 |
| **Variables and types** | Programming: Principles and Practice | 2.5 Assignment and initialization | 49 to 52 |
| | Programming: Principles and Practice | 2.8 to 2.10 Type safety, conversions, type deduction with auto | 56 to 60 |
| | Effective Modern C++ | Item 7, Distinguish () and {} when creating objects | 67 to 76 |
| **Operators and control flow** | C++ Primer | 4.2 to 4.3 Arithmetic, logical and relational operators | 168 to 172 |
| | C++ Primer | 4.4 to 4.5 Assignment, increment and decrement | 173 to 178 |
| | C++ Primer | 5.3 to 5.5 Conditional, iterative and jump statements | 203 to 221 |
| **Functions and passing arguments** | C++ Primer | 6.1 Function basics | 231 to 236 |
| | C++ Primer | 6.2 Argument passing | 237 to 250 |
| | C++ Primer | 6.3 Return types and the return statement | 251 to 258 |
| | C++ Primer | 6.4 Overloaded functions | 259 to 264 |
| | C++ Primer | 6.6 Function matching | 271 to 275 |
| | Programming: Principles and Practice | 3.5 to 3.6 Functions; vector introduced | 81 to 89 |
| | Programming: Principles and Practice | 7.2 to 7.3 Declarations and definitions; scope | 194 to 202 |
| | Programming: Principles and Practice | 7.4 Function call and return (pass by value, by reference, by const reference) | 203 to 218 |